Project Report: VAce_QuizMaster

👤 Student Details

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- **Submission Date:** March 31, 2025

📌 Project Overview

- **Title:** VAce_QuizMaster
- **Objective:**
  The project is a full-stack web application titled **VAce_QuizMaster**, designed as an interactive quiz platform. It supports two distinct user roles:
  - **Admin**: Manages users, subjects, chapters, quizzes, questions, and tracks analytics.
  - **User**: Registers, logs in, attempts quizzes, and reviews their quiz history and scores.

❓ Development Approach

📋 Requirement Analysis

- Identified core functional modules and roles
- Defined user flows and interface expectations
- Focused on modular, maintainable design with role-based access

🔧 Tech Stack Finalization

- **Backend**: Flask (Python)
- **Templating**: Jinja2
- **Frontend**: HTML5, CSS3, Bootstrap 5
- **Database**: SQLite (ORM via SQLAlchemy)
- **Security**: Werkzeug for password hashing and role verification

🗄 Database Design

Modeled the following entities:

- User, Subject, Chapter, Quiz, Question, Option, Score
- Applied **1:N relationships** consistently using foreign keys
- Password hashing, role enforcement, and cascade rules handled in models

🔙 Backend Development (Flask)

- Routes grouped by function: authentication, admin, user
- Admin features include:
  - CRUD for users, subjects, chapters, quizzes, questions
  - Real-time analytics and CSV export
- User features include:
  - Quiz dashboard with scores and history
  - Quiz attempt interface with validation
- Decorators for role-based access: login_required, admin_required

🎨 Frontend Development (Jinja2 + Bootstrap)

- Shared base.html layout with role-based navbar
- Clean UI with flash messages, modals, tables, and cards
- Admin dashboard displays latest users/quizzes and statistics
- Charts powered by Chart.js for analytics

🧪 Integration & Testing

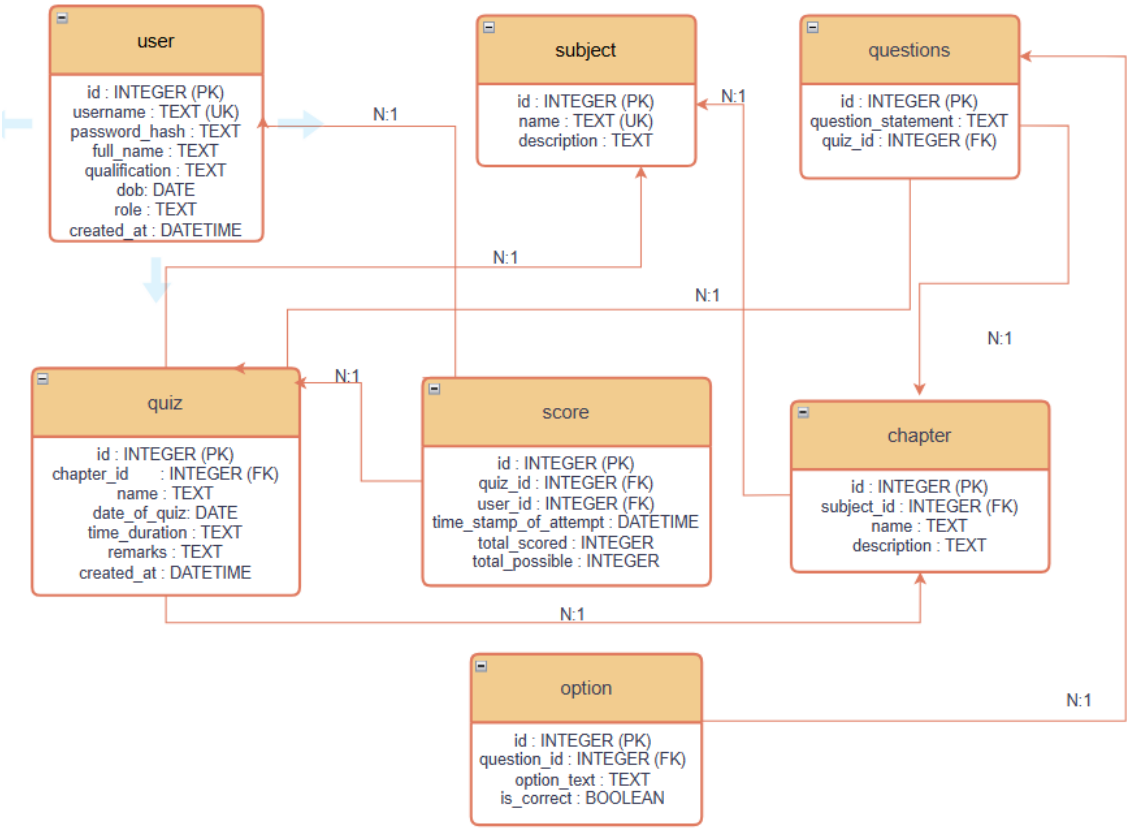
- Manual testing for:
  - Quiz creation and attempt logic
  - Role-based access restrictions
  - Flash messages and form validation
- UI responsiveness verified on multiple screen sizes
- Verified relationships and cascade operations on delete

⚙ Tools & Libraries

- **Flask** – Web framework
- **Flask-SQLAlchemy** – ORM
- **Werkzeug** – Password security
- **Jinja2** – Templating engine
- **Bootstrap 5** – UI components

- **Chart.js** – Data visualization
- **Standard Libraries** – datetime, os, csv, functools, io

ER Diagram & Summary



Entity	Attributes	Relationships
User	id (PK), username (UK), password_hash, full_name, qualification, dob, role, created_at	1:N → Score
Subject	id (PK), name (UK), description	1:N → Chapter
Chapter	id (PK), subject_id (FK), name, description	1:N → Quiz
Quiz	id (PK), chapter_id (FK), name, date_of_quiz, time_duration, remarks, created_at	1:N → Question, Score
Question	id (PK), quiz_id (FK), question_statement	1:N → Option

Option	id (PK), question_id (FK), option_text, is_correct	Belongs to Question
Score	id (PK), quiz_id (FK), user_id (FK), time_stamp_of_attempt, total_scored, total_possible	Belongs to Quiz and User

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🎬 Presentation Video

Drive Link:
https://drive.google.com/drive/folders/1B_2H2BmWDGezvLKohi3xJhkJmol3ZACt?usp=sharing